

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Problem Formulation and Representations

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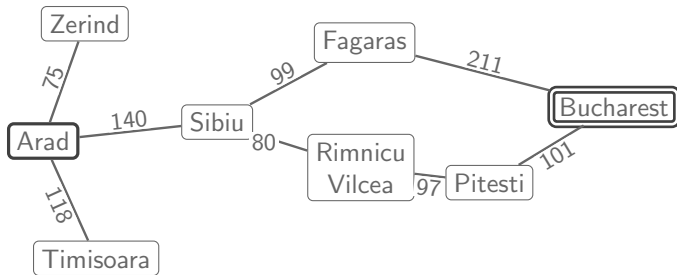
University of Arkansas

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What Must the Agent Represent?

Situation

The agent is in Arad. Its flight leaves Bucharest tomorrow, and a road map is available.

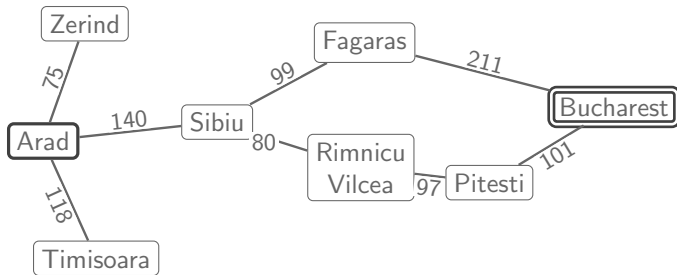


Before choosing a road, what must the agent represent?

What Must the Agent Represent?

Situation

The agent is in Arad. Its flight leaves Bucharest tomorrow, and a road map is available.



Before choosing a road, what must the agent represent?

A state space, an initial state, goal states, applicable actions, a transition model, and action costs.

The Problem-Solving Cycle



Two Different Commitments

A problem-solving agent uses search to look ahead for a path to a goal. **Goal formulation** selects an outcome; **problem formulation** decides which states and actions matter.

The Problem-Solving Cycle



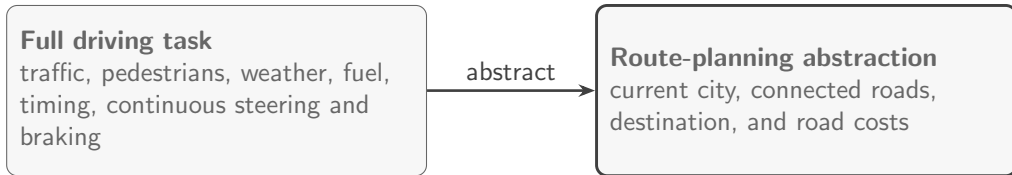
Two Different Commitments

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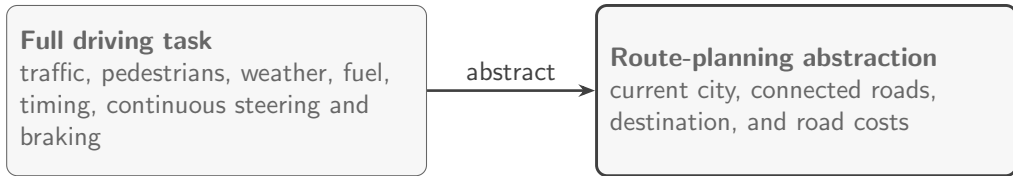
Open-Loop Execution

Under the stated classical assumptions and a correct model, a solution is a fixed action sequence and may be executed without consulting new percepts.

Driving vs. Route Planning



Driving vs. Route Planning



Classical Search Setting

Fully observable; known transition model; single-agent; deterministic; static; discrete; sequential.

These are modeling assumptions, not properties of every AI task.

Search Problems: Six Components

Component	Notation	Question answered
State space	$\mathcal{S}$	Which abstract states are possible?
Initial state	s_0	Where does planning begin?
Actions	$\text{ACTIONS}(s)$	What can the agent do in state s ?
Transition model	$\text{RESULT}(s, a)$	What state results from doing a in s ?
Goal states or test	G or $\text{IS-GOAL}(s)$	When is the task complete?
Action cost	$c(s, a, s')$	What does one transition cost?

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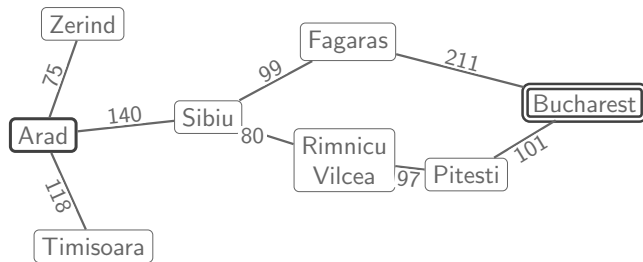
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Path cost accumulates action costs:

$$g = \sum_{i=0}^{n-1} c(s_i, a_i, s_{i+1}).$$

A goal test can recognize one or many goal states; action costs are assumed positive.

Romania Route Planning



Element

Romania model

State space

Cities; each state records the current city

Initial

Arad

Actions

Drive to an adjacent city

Result

Arrival in that city

Goal

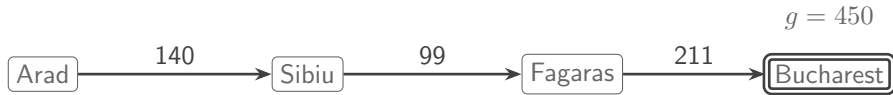
Bucharest

Cost

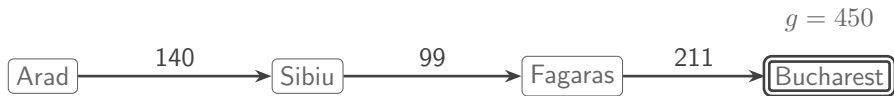
Road distance

Each road gives two directed drive actions; the map is a model, not an algorithm.

Solutions and Path Costs



Solutions and Path Costs



Solution

A path—an action sequence from the initial state—to a goal state.

Path Cost

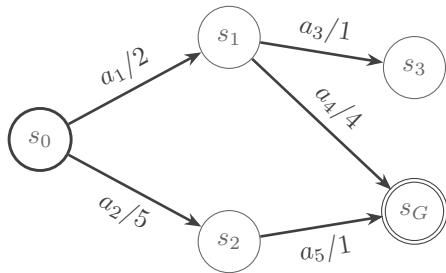
The sum of its action costs.

Optimal Solution

A solution with minimum path cost.

Optimality depends on the chosen cost: distance, time, and money can disagree.

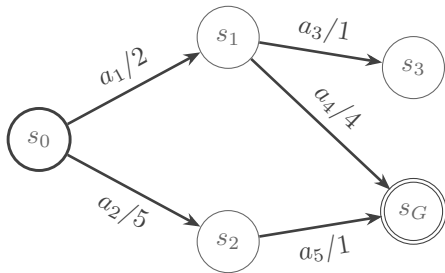
The State-Space Graph



Edge label: action / cost

Element	Problem meaning
Vertex	Abstract state
Directed edge	Applicable action
Endpoint	$\text{RESULT}(s, a)$
Edge weight	Action cost

The State-Space Graph

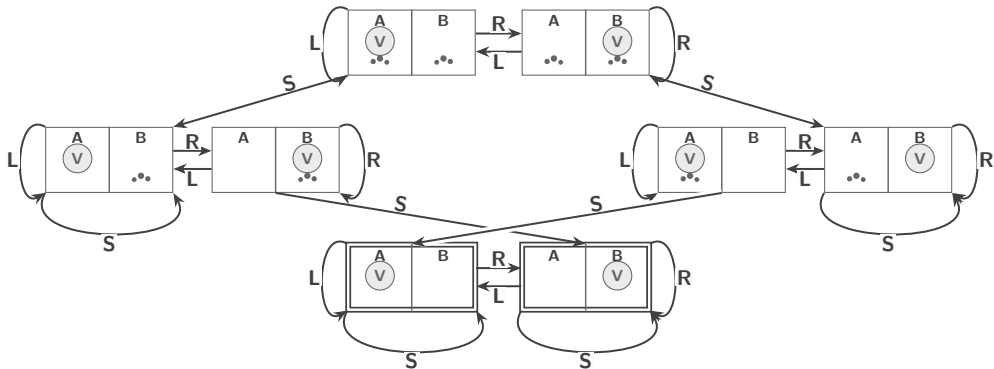


Edge label: action / cost

Problem formulation defines the graph; search decides how to explore it.

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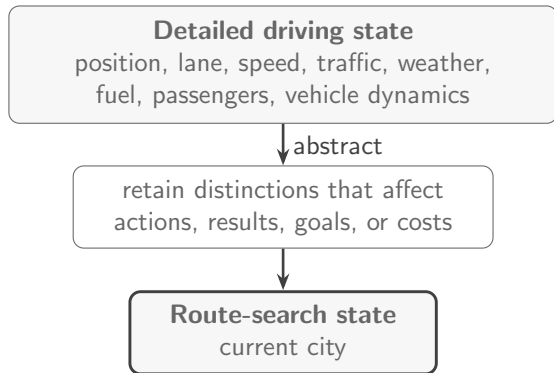
Vacuum-World State Space



State key: V = agent, dots = dirt. **Actions:** L = Left, R = Right, S = Suck; NoOp is not part of this formulation.

Loops mean no state change; double outlines mark goal states; every action costs 1.

Search-State Abstraction



Valid

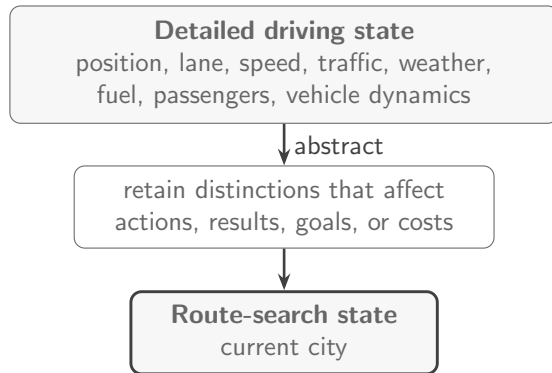
Every abstract solution can be elaborated into a solution in the more detailed world.

Useful

Executing an abstract step is easier than solving the original task from scratch.

Generic search treats each complete state as a whole; it does not exploit internal structure.

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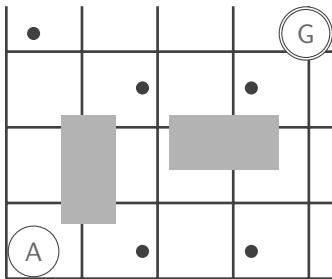
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Remove detail while preserving valid solutions and easy-to-execute abstract actions.

Same World, Different State



One grid world; two goals.

N : traversable cells; D_0 : initial dots

Reach the Goal Cell

State: (x, y)

A position is enough to predict moves and test the goal.

State-space upper bound: N .

Visit Every Dot

State: $((x, y), D)$, where $D \subseteq D_0$ is the set of remaining dots.

The same position can leave different work unfinished.

State-space upper bound: $N 2^{|D_0|}$.

Changing the goal can change what the state must remember.

Discussion: Scale and Abstraction

For visit-every-dot, $|\mathcal{S}| \leq N 2^k$, where $k = |D_0|$.

Larger Instance

What happens to the bound if the grid size N doubles? What if the problem has ten additional dots?

Retain More Detail

What if every state also records facing direction—north, east, south, or west—and fuel with F values? When is that detail necessary?

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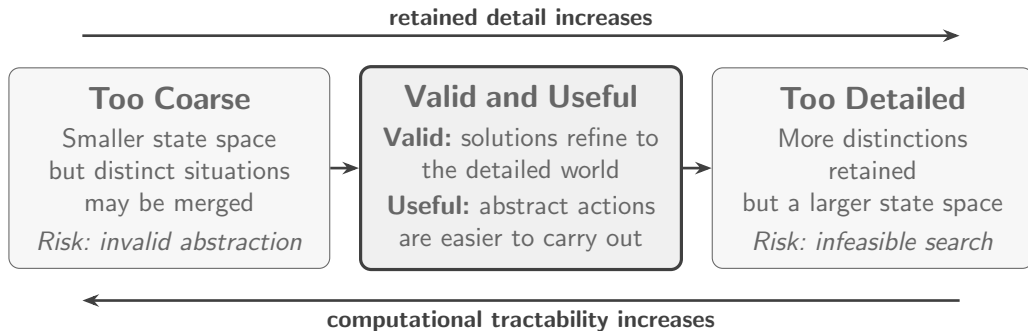
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We Can—But at a Computational Price

Doubling N can double the upper bound; ten more dots can multiply it by $2^{10} = 1024$; four facing directions and F fuel levels can multiply it by another $4F$. Time and memory can become infeasible.

Which details are worth their computational cost?

The Abstraction Tradeoff



Too Coarse in the Dot World

Using only (x, y) does not determine which dots remain or whether the visit-every-dot goal holds.

Too Detailed When Irrelevant

Adding four facing directions and F fuel values multiplies the bound by 4^F when neither changes actions, results, goals, or costs.

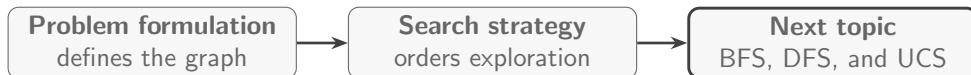
Use the coarsest abstraction that remains valid and useful.

Formulation Before Search

Audit question	Why it matters
Does the state retain enough information?	Actions, results, goal status, and costs must follow from the state.
Are actions at the right level?	Each action should be abstract enough to plan with and concrete enough to execute.
Does the goal match the objective?	A convenient proxy can lead to the wrong destination.
Do costs encode the preference?	Distance, time, risk, and money can rank paths differently.
Has irrelevant detail been removed?	Smaller state spaces can make the same task far easier to solve.

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Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.